

LARISSA, Greece

WMO: 166480

Lat: 39.650N

Lon: 22.466E

Elev: 74

StdP: 100.45

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.8	-2.9	-9.3	1.7	0.1	-7.0	2.1	1.0	10.3	5.7	9.0	6.3	0.7	300	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	36.9	20.8	35.2	20.6	34.0	20.3	22.7	32.5	22.0	31.6	21.3	30.5	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	14.9	24.9	19.2	14.1	24.5	18.5	13.5	24.2	67.4	32.3	64.6	31.6	62.3	30.6	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.4	6.3	-8.4	40.6	3.8	2.0	-11.2	42.1	-13.4	43.3	-15.6	44.5	-18.3	46.0		
			-8.9	24.7	3.7	1.5	-11.5	25.8	-13.7	26.7	-15.7	27.5	-18.4	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	5.4	7.1	10.0	13.9	19.4	24.3	26.6	26.1	21.8	16.6	11.2	6.4
	DBStd	8.10	3.48	3.36	3.31	2.96	2.93	2.76	2.20	2.00	2.83	3.10	3.48	3.75
	HDD10.0	429	146	89	41	5	0	0	0	0	0	1	27	121
	HDD18.3	1794	401	313	259	137	23	0	0	0	5	71	216	369
	CDD10.0	2537	3	9	41	121	290	429	513	498	355	205	62	10
	CDD18.3	860	0	0	1	2	55	180	255	240	110	17	1	0
	CDH23.3	10148	0	0	13	68	703	2161	3184	2808	1037	168	6	0
	CDH26.7	4926	0	0	4	10	227	1064	1714	1466	408	34	0	0
Wind	WSAvg	1.7	1.2	1.7	1.9	1.9	2.0	2.2	2.1	2.0	1.8	1.2	1.0	1.1
Precipitation	PrecAvg	425	34	35	39	31	38	25	19	16	32	48	54	52
	PrecMax	704	105	122	116	123	132	106	132	70	240	127	169	156
	PrecMin	211	1	5	9	2	1	0	0	0	0	0	2	2
	PrecStd	92	25	25	23	26	29	25	26	18	41	35	39	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	20.2	24.4	28.1	33.7	38.2	40.2	38.2	35.2	30.0	24.0	18.9
		MCWB	12.4	12.5	13.8	16.4	18.6	20.9	21.7	21.0	19.5	18.5	15.7	13.9
	2%	DB	14.8	17.6	21.8	25.2	31.0	35.9	37.1	36.2	32.9	27.1	20.9	16.0
		MCWB	10.3	11.4	13.0	15.5	17.8	20.3	20.7	20.9	19.4	17.4	15.2	12.5
	5%	DB	12.9	15.7	19.7	23.0	29.1	34.0	35.2	34.8	31.0	25.0	18.8	13.8
		MCWB	9.4	10.8	12.4	14.6	17.3	20.0	20.6	20.9	18.9	16.9	14.2	11.0
	10%	DB	11.0	13.6	17.2	21.1	27.3	32.1	33.9	33.2	29.0	23.0	16.9	12.0
		MCWB	8.5	9.6	11.3	13.7	16.7	19.5	20.3	20.5	18.5	16.2	13.3	9.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	13.9	15.0	17.9	20.1	23.4	23.4	24.2	21.7	20.4	17.2	15.0
		MCDB	16.4	17.6	22.3	25.6	29.3	34.2	34.0	32.9	28.6	26.1	20.9	16.9
	2%	WB	11.1	12.2	13.7	16.3	18.9	22.0	22.5	22.9	20.8	18.9	15.9	13.0
		MCDB	13.9	16.2	20.2	23.3	28.2	32.8	32.6	32.3	28.2	23.9	19.3	15.4
	5%	WB	9.9	11.1	12.6	15.3	18.2	21.1	21.8	22.1	20.1	18.0	14.9	11.5
		MCDB	12.0	14.7	18.5	21.6	26.7	30.8	31.7	31.2	27.3	22.4	17.7	13.4
	10%	WB	8.8	10.0	11.6	14.2	17.5	20.3	21.2	21.4	19.5	17.2	13.9	10.1
		MCDB	10.6	13.1	16.7	19.7	25.2	29.4	31.0	30.0	26.3	21.3	16.3	11.6
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.3	11.9	12.7	14.1	14.5	14.2	14.2	13.1	11.4	9.9	8.7
		MCDBR	13.0	14.6	16.7	16.8	17.3	17.2	16.9	16.7	16.4	15.3	12.6	11.5
		MCWBR	9.0	9.1	9.2	8.4	6.5	5.4	5.3	5.5	5.9	7.0	7.4	8.0
	5% WB	MCDBR	11.1	13.1	15.2	14.9	15.3	15.3	14.8	14.3	13.7	12.0	10.7	9.7
		MCWBR	8.1	8.4	8.8	7.9	6.2	5.5	5.1	5.4	5.4	5.9	6.7	7.1
Clear-Sky Solar Irradiance	taub	0.360	0.390	0.437	0.490	0.479	0.468	0.475	0.486	0.464	0.428	0.393	0.367	
	taud	2.360	2.231	2.099	1.975	2.073	2.151	2.140	2.101	2.138	2.218	2.269	2.320	
	Ebn at Noon	791	812	810	791	810	818	807	786	775	761	738	745	
	Edh at Noon	89	115	146	176	163	151	151	153	139	116	97	86	
All-Sky Solar Radiation	RadAvg	1.87	2.63	3.85	5.08	6.11	6.97	7.24	6.43	4.70	3.14	2.11	1.64	
	RadStd	0.29	0.33	0.39	0.52	0.31	0.42	0.30	0.29	0.35	0.36	0.25	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	+0.38	N/A	N/A	+0.72	N/A	N/A	N/A	-109	N/A	N/A
		+0.36	N/A	N/A	+0.66	N/A	N/A	N/A	-89	+102	N/A

Nomenclature: See separate page